

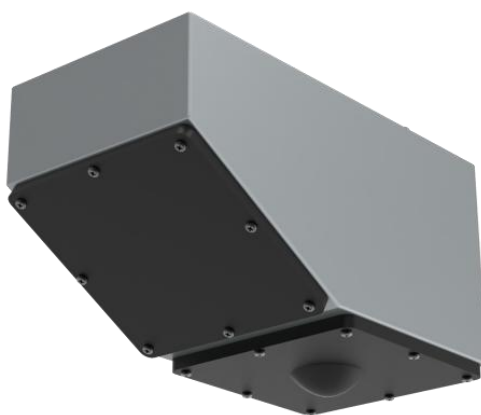


RS-RAD-N01-3

Radar Flow-meter Sensor

User Manual

Document version:V2.1





Declaration

1. This user manual is copyrighted by Shandong Renke Control Technology Co., LTD. (hereinafter referred to as "the Company"), without written permission, no part of this user manual can be copied, translated, stored in a database or retrieval system, and can not be disseminated by any means such as electronic, rephotographing, or audio recording.

2. Thank you for using Shandong Renke series products. To enable you to use our company's products better and reduce product malfunctions caused by improper use, please read this manual carefully before use and follow the recommended usage methods. If the user does not use the equipment in accordance with this manual or removes, disassembles or replaces the internal components of the equipment without authorization, our company shall not be held responsible for any losses caused thereby.

3. Our company adheres to the concept of advancing science and technology, and continuously commits to product improvement and technological innovation. Therefore, our company reserves the right to make any product improvements without prior notice. When using this manual, please confirm that it is the valid version.

Please keep this user manual properly for easy reference and assistance when you need it in the future.

Shandong Renke Control Technology Co.,Ltd



Catalog

1. Product introduction	4
1.1 Product overview	4
1.2 Function features	4
1.3 Main technical parameter	4
1.4 Product model	5
2. Equipment instruction	5
2.1 Radar flow-meter size	5
2.2 Equipment inspection before installation	5
2.3 Installation instruction	5
2.4 Environment selection	6
2.5 Antenna beam range	9
2.6 Effect of mounting height on measurement	10
2.7 Installation precautions	10
2.8 Interface specification	11
3. Configuration software installation and use	11
3.1 Software selection	11
3.1.1 Parameter settings	11
4. Communication protocol	15
4.1 Basic communication parameters	15
4.2 Definition of data frame format	16
4.3 Register address	16
4.4 Communication protocol examples and explanations	18
4.5 Flow rate calculation related parameters	18
5. Common problems and solutions	19
5.1 The device cannot be connected to a PLC or computer	19
5.2 Working principle	19
6. Precautions	20
7. Warranty Statement	20
8. Contact information	21
9. Document history	21



1. Product introduction

1.1 Product overview

This is a kind of radar flow-meter series product, which can be used with our company's water and rain situation poles and host. 485 communication, standard ModBus-RTU communication protocol, with a maximum communication distance of 2000 meters. The product is powered by a 10~30V wide voltage power supply, be mainly used in situations where flow and water level monitoring is required, such as canals, rivers and so on.

This sensor can detect water flow velocity and water level height. The flow meter uses radar technology to achieve a simple and fast non-contact surface velocity measurement. The non-contact radar flow measuring system is not corroded by sewage and not affected by sediment when measuring flow velocity. Collect the fluid surface flow velocity, and get the average flow velocity of the overflow section through the model calculation, then the water level and section information measured by the water level meter are used to calculate the overflow section area, so as to obtain the flow by the formula $\text{flow} = \text{average flow velocity} \times \text{overflow section area} \times \text{correction coefficient}$, in which the correction coefficient is calculated according to the standard value obtained in the actual measurement environment.

1.2 Function features

- Non-contact measurement, protection from sediment and easy maintenance
- Can detect water velocity, water level height, water volume and flow
- 485 communication, standard ModBus-RTU communication protocol
- 10~30V DC wide voltage power supply
- Water-proof design which can be used outdoor

1.3 Main technical parameter

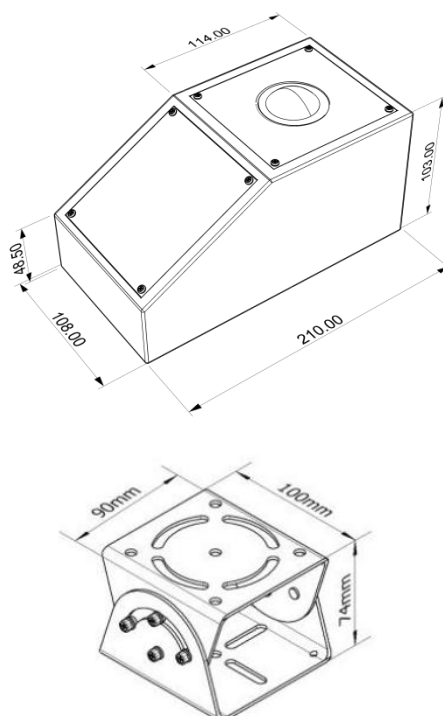
Power supply	DC 10-30V
Maximum power consumption	1.3W
Transmitter components temperature and humidity resistance	-40°C~+80°C, 0%RH~95%RH(non-condensation)
Velocity measurement transmitting frequency	24.00GHz
Velocity measurement range	0.1 ~20m/s
Velocity measurement precision	±2%
Velocity resolution	0.01m/s
Distance measurement transmitting frequency	77GHz~79GHz
Water level distance measurement range	0.15m ~ 65m(Maximum range when measuring water level only)
Distance measurement precision	±1mm
Distance measurement resolution	1mm
Use height	0~5m
Waterproof grade	IP67

1.4 Product model

Code model						
RS-					Company code	
	RAD-				Radar detection	
		N01-			RS485(ModBus protocol)	
				3-		Radar flow-meter
					EX	Fixed code

2. Equipment instruction

2.1 Radar flow-meter size



2.2 Equipment inspection before installation

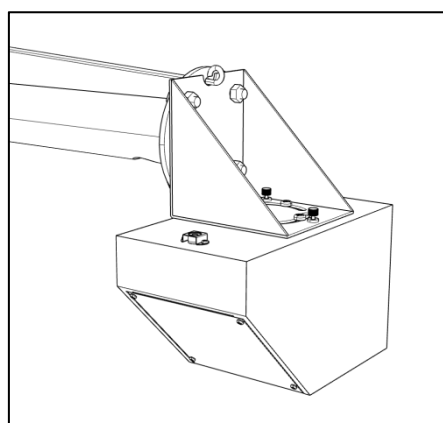
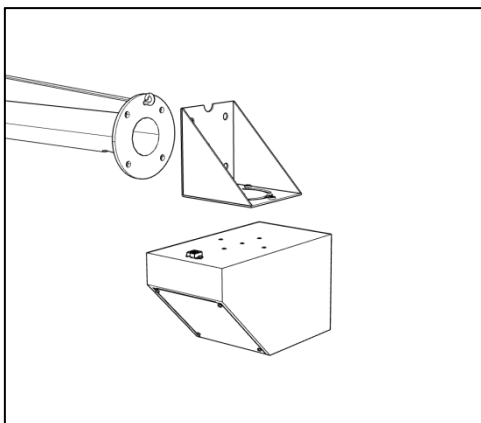
Equipment list:

- One radar flow-meter device
- Water-proof pair cable male cable 5m
- Mounting bracket
- Mounting screws
- Certificate of conformity, warranty card

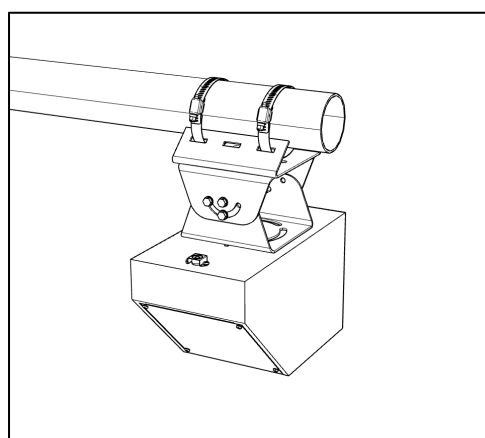
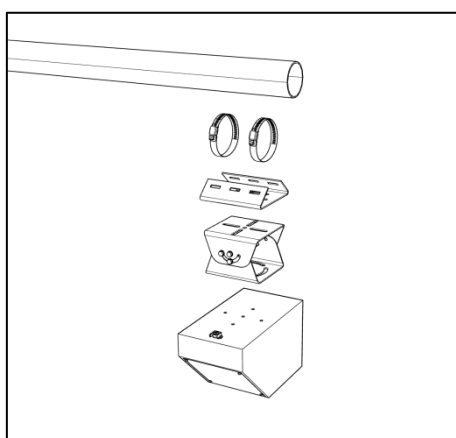
2.3 Installation instruction

Bracket installation instruction

Water and rainfall standing pole installation

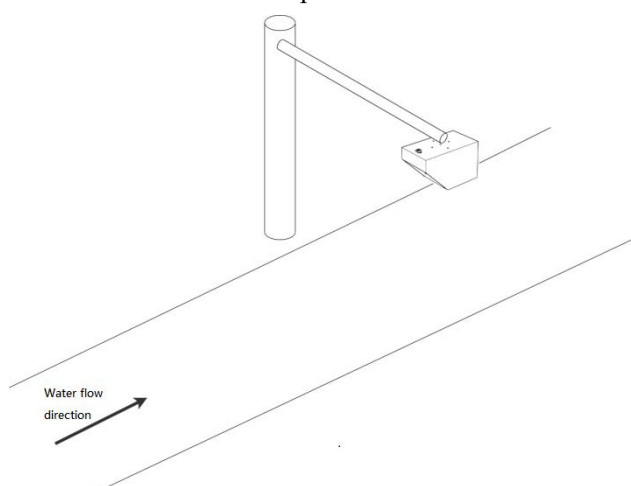


Conventional standing pole installation



Installation environment instruction

Install the inclined side of the radar flow-meter parallel to the flow direction in the middle of the channel or river or as close to the center as possible.



2.4 Environment selection

The selection of measurement channel has a direct relationship with the measurement flow precision. In order to obtain better measurement results, the following conditions should be met as far as possible:

1. There are no large stones blocking water, no large eddies, turbulence, or other phenomena in the test canal.
2. The test canal should be straight, stable, and with concentrated water flow.
3. The test canal needs to be hardened, and the measurement section should be neat.
4. The test canal should be kept smooth to prevent the accumulation of floating objects.

【Note】 When radar waves are incident on the water surface at an oblique angle, the calm water surface will reflect the electromagnetic waves in the opposite direction of the radar, resulting in a weak reflected signal received by the radar. This phenomenon has no direct correlation with the magnitude of the flow velocity. The key is that the radar must receive sufficient energy to perform data measurement.

As shown in Figures 1 and 2 below, when the water surface is stationary, the reflected energy is weak, which may lead to inaccurate measurement results or failure to measure. Conversely, when there are disorderly ripples on the water surface or the ripples are not obvious, the stability of the data will be affected, thereby resulting in significant differences in the measured water flow values.



Figure 1



Figure 2

The water surface flow regime with fish-scale-like ripples exhibits excellent reflected signal characteristics and can effectively reflect signals, please refer to Figures 3 and 4 below for details.



Figure 3



Figure 4

The following scenarios should be avoided as much as possible:



The area where vortices form in the river channel



Areas with complex river flow patterns



The location of the pipe outlet in the river channel



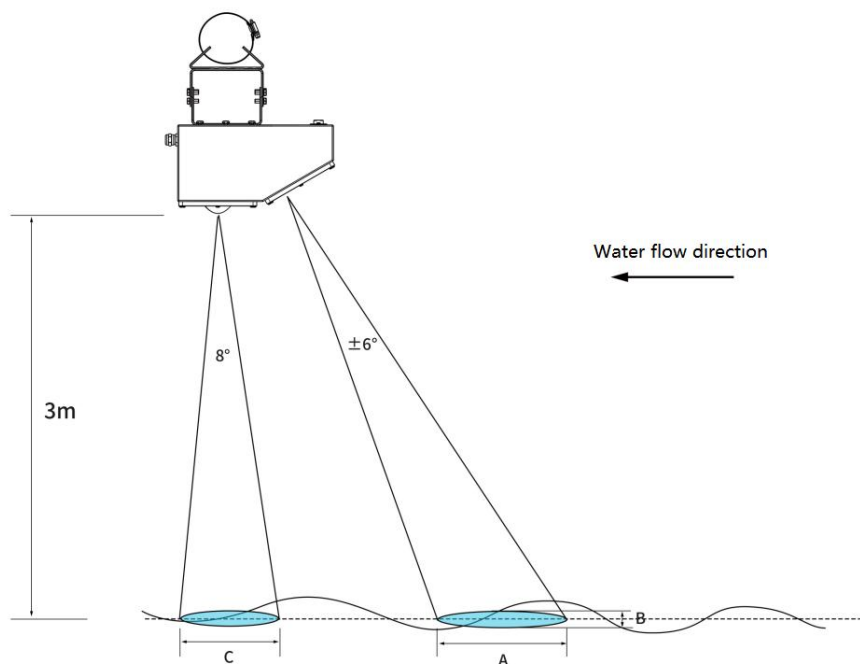
The location of the water outlet in the rapid current area of the river course



The rapid currents section of the river with inclined DAMS

2.5 Antenna beam range

The flow-meter integrates a radar water level gauge and a radar flow velocity gauge, and the antenna beam angle of the radar water level gauge is 8° , radar flow velocity beam angle is $\pm 6^\circ \pm 12^\circ$. When the water level gauge illuminates the water surface, the illuminated area resembles a circle, and when the flow velocity gauge illuminates the water surface, the illuminated area resembles an elliptical area, as shown in the figure. Accurately understanding the irradiation range of radar waves helps to select suitable locations for installation and avoid some easily disturbed scenes, such as the swaying branches on both sides of the river, floating debris in the water and so on.



Install 3 meters radar flow-meter antenna irradiation area

The boundary of the radar illuminated water surface area is proportional to the installation height. The following table shows the A,B,D parameter values (A, B, D) of the water level meter and velocity meter beam irradiated by the water surface when the installation height is 3 meters. The actual installation height (unit meter) multiplied by the following values is the actual corresponding parameter.

Category	Length(m)
Velocity meter A	About 3.46
Velocity meter B	About 7
Water level meter diameter C	About 0.32

Parameter value of antenna beam irradiation surface

2.6 Effect of mounting height on measurement

Under the same conditions, the higher the installation height, the weaker the echo, the worse the signal quality, especially for the scene with low water velocity, the ripple is small, more difficult to measure, and the radar wave irradiation area will be larger at the same time, it is possible that the beam will illuminate the canal bank, will be affected by the moving target on the bank. If the installation is too low, it is not conducive to anti-theft protection, so the recommended installation height range is 3-4 meters for the standing pole installation.

2.7 Installation precautions

- 1) When install the flow-meter, water level meter and velocity meter radar should not be shielded, otherwise the measurement accuracy will be affected.

- 2) When installing the flow meter, ensure that the upper surface of the shell is horizontal and installed in the middle of the channel.
- 3) It is recommended that the beam of the current meter face the direction of the incoming water, as Figure 3-3 shows, and the horizontal angle with the water flow direction is 0 degrees.
- 4) The current meter will be only affected by the dynamic target, and when the channel is hardened and there are no weeds or trees, even if the beam shines on both sides of the channel, it will not affect the flow measurement.

2.8 Interface specification

Wide voltage power input 10~30V can be used. When wiring 485 signal cable, pay attention that cable A/B can not be connected inversely, and address of multiple devices on the bus should not be conflicted with each other.

	Cable color	Specification
Power supply	Brown	V+
	Black	V-
Communication	Yellow(green)	485-A
	Blue	485-B

3. Configuration software installation and use

3.1 Software selection

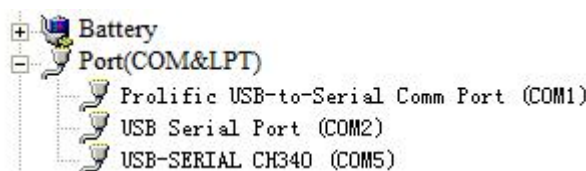
Open the data package, select "Debugging software" -- "485 parameter configuration software",



find **VS01.exe** and open it.

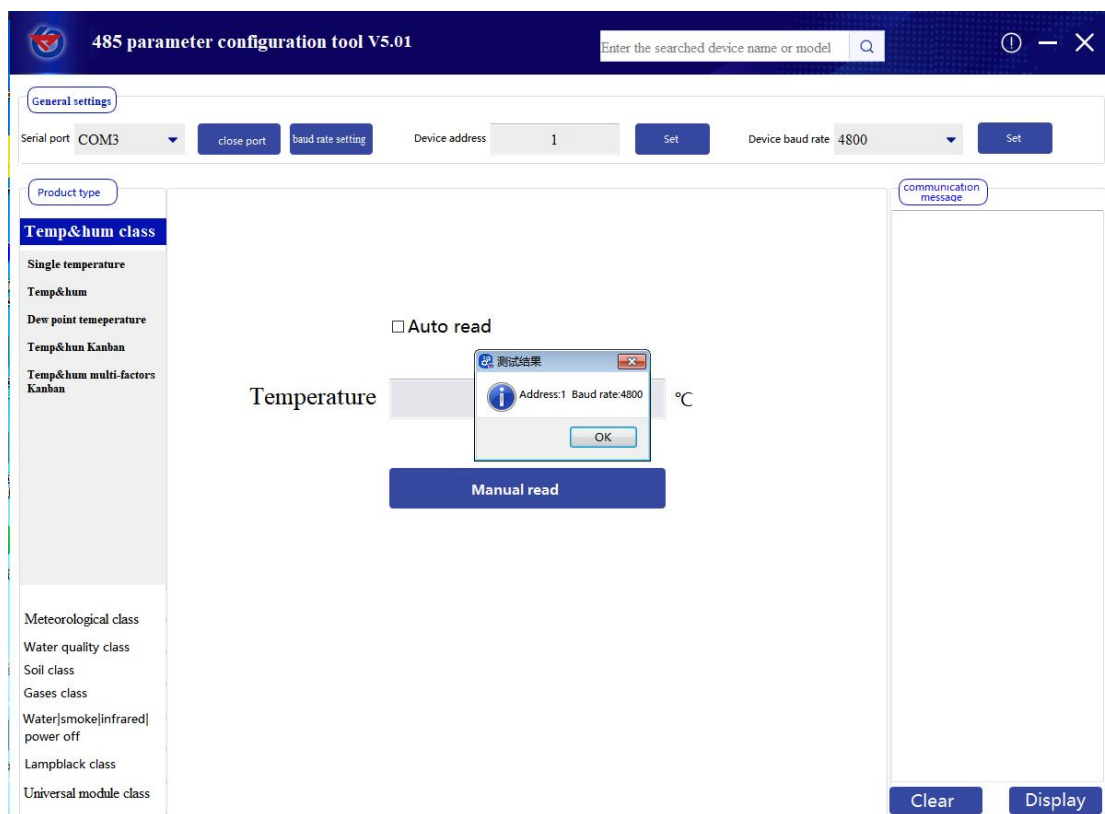
3.1.1 Parameter settings

- ① Select the correct COM port (see COM ports in "My Computer - Properties - Device Manager - Ports"). The following figure lists the driver names of several different 485 converters.



- ② Connect only one device and power it on. Click the test baud rate of the software. The software will test the baud rate and address of the current device. Default baud rate is 4800bit/s, default address is 0x01.
- ③ You can modify the address and baud rate as the required of use, and query the current function status of the device.

- ④ If the test is not successful, check the device wiring and the 485 driver installation status.



3.2 Mobile phone software selection(Android)

The device supports Bluetooth configuration. You need to download the configuration app "Tap-to-Connect Bluetooth Configuration" on your mobile phone. The app can be obtained by contacting the staff of our company or scanning the QR code with Mobile QQ.



Search the connected device

- (1) After finish download, open Bluetooth, the opened APP software interface as below, click "Connect device", enter the device scanning page.

3.2.1 Search the connected device

- (1)The device support Bluetooth configuration after power on.
(2)Open the Bluetooth function of mobile phone, then click the APP that installed just now to

enter main page, select Bluetooth device.



(3)Click “Connect device” enter the page that shows the scanned device.



(4)Click “Start scanning” to search the device that need to be configured, device name is 3066+ 8-bit address.(total 12-bit address)

【Note】 Assume you searched a device which named 306612345678, the address of the device is 306612345678.

(5)Click the password input box, input device password (default password is 12345678), enter the device configuration page.



3.3 Real-time data instruction



The real-time data be used to check the device data collector and status judgement.

Water Volume: All water volume recorded from the time of clearing the water volume to the current time.

Flow velocity: Current surface velocity of the measured water surface.

Water level: The current measured water surface height(calculated by subtracting the empty height from the measuring range)

instantaneous flow rate: The flow rate of current measured water surface(the cross-section of the measured water needs to be set in advance)

3.4 Configuration instruction



The parameters that not marked mean they are not available for this device, please do not modify them.

ModBus address: The 485 address of current device.

Baud rate: The communication signal transmitting rate of current device(need to be consistent with collecting host).

Parity check method: The check method to detecting data(no check, odd parity check ,even parity check can be set).

Device address: The unique address of device, which can be used to mark the device identity.

Section shape: The cross-section shape of current measured river or canal.(square or trapezoid can be set)

Section dimension: Please see “4.5 Flow Calculating Related Parameters” for details.

Power on time recording: The time that start to record water volume.

Host clear stored data: Clear the store water volume in the current device, and restart recording.

Water level range: The overall height from device to water bottom.

4. Communication protocol

4.1 Basic communication parameters



Encode	8-bit binary
Data bits	8bit
Parity check bit	-
Stop bit	1bit
Error verification	CRC(Cyclic Redundant code)
Baud rate	1200bit/s, 2400bit/s, 4800bit/s, 9600 bit/s, 19200bit/s, 38400bit/s, 57600bit/s, 115200bit/s can be set, factory default is 9600 bit/s

4.2 Definition of data frame format

Adopting ModBus RTU communication protocol, the format is as follows:

Time for initial structure ≥ 4 bytes of time

Address code=1 byte

Function code=1 byte

Data area=N bytes

Error check=16 bit CRC code

Time to end structure ≥ 4 bytes of time

Address code: The address of the transmitter, which is unique in the communication network (factory default 0x01).

Function code: The instruction function indication issued by the host.

Data area: The data area is specific communication data, please note that the 16bits data high byte comes first!

CRC code: Two byte check digit.

Host inquiry frame structure:

Address code	Function code	Register Start Address	Register length	Check code low bit	Check code high bit
1 byte	1 byte	2 byte	2 byte	1 byte	1 byte

Slave response frame structure:

Address code	Function code	Effective Bytes number	Data Zone 1	Second data area	Nth data area	Check Code
1 byte	1 byte	1 byte	2 byte	2 byte	2 byte	2 byte

4.3 Register address

Register address	Configuration address	Illustrate	Function code(hex)	Data type	Description
03E8 H	41001	Water volume high 16-bit	03/04	INT32U	Unit m ³
03E9H	41002	Water volume low 16-bit	03/04		
03EAH	41003	Instantaneous	03/04	INT16U	Unit: m ³ /s



		flow rate value			Enlarge 1000 times to upload(maximum can be displayed to 65.535m ³ /s)
0401H	41026	Water level empty height	03/04	INT16U	Unit mm
0437H	41080	Instantaneous flow rate value	03/04	INT16U	Unit: m ³ /s
03EBH	41004	Current water level value	03/04	INT16U	Unit: mm
03ECH	41005	Current flow rate value	03/04	INT16U	Unit: cm/s
0422H	41059	Water level range	03/04/06/10	INT16U	Unit: mm 0-65535mm
03F4H	41013	Address	03/04/06/10	INT16U	1-254
03F6H	41015	Baud rate	03/04/06/10	INT16U	0 represent 2400 1 represent 4800 2 represent 9600
					3 represent 19200 4 represent 38400 5 represent 57600
					6 represent 115200 7 represent 1200
0412H	41043	Section type	03/04/06/10	INT16U	1: trapezoid, 2: rectangle(suggest to use configuration software)
0413H	41044	Section dimension 1	03/04/06/10	INT16U	Unit: mm (suggest to use configuration software)
0414H	41045	Section dimension 2	03/04/06/10	INT16U	Unit: mm(suggest to use configuration software)
0415H	41046	Section dimension 3	03/04/06/10	INT16U	Unit: mm(suggest to use configuration software)
0417H	41048	Water level jump threshold	03/04/06/10	INT16U	Unit: mm
0418H	41049	Flow rate jump threshold	03/04/06/10	INT16U	Unit: mm
0430H	41073	Water volume	03/04	INT32U	Current water volume



		record time high bit			value start record time
0431H	41074	Water volume record time low bit	03/04		
03E9H	41002	Clear current water volume	06/10	INT16U	Write 0x5A5A(restart record current water volume value)
0423H	41060	Time revision enable	06/10	INT16U	Write to time register when write 1
0439H	41082	Flow velocity direction	03/04	INT16U	Against the stream: 1 With the stream: 0
0424H	41061	year	03/04/06/10	INT16U	Integer
0425H	41062	month	03/04/06/10	INT16U	Integer
0426H	41063	day	03/04/06/10	INT16U	Integer
0427H	41064	hour	03/04/06/10	INT16U	Integer
0428H	41065	minute	03/04/06/10	INT16U	Integer
0429H	41066	second	03/04/06/10	INT16U	Integer

4.4 Communication protocol examples and explanations

4.4.1 Read device water level value

Inquiry frame

Address code	Function code	Start address	Data length	Check code low bit	Check code high bit
0x01	0x03	0x03 0xEB	0x00 0x01	0xF4	0x7A

Response frame (for example, the reading water level is 1.064m)

Address code	Function code	Return the valid byte number	Data area	Check code low bit	Check code high bit
0x01	0x03	0x02	0x04 0x28	0xBA	0x9A

4.4.2 Read device flow rate value

Inquiry frame

Address code	Function code	Start address	Data length	Check code low bit	Check code high bit
0x01	0x03	0x03 0xEA	0x00 0x01	0xA5	0xBA

Response frame (for example the reading instantaneous flow value is 1.011m³/ s)

Address code	Function code	Valid byte number	Data area	Check code low bit	Check code high bit
0x01	0x03	0x02	0x03 0xF3	0xF8	0xF1

4.5 Flow rate calculation related parameters



4.5.1 Section parameters

Section parameters are mainly used to calculate section area. Section parameters include section type and section size 1, section size 2, and section size 3.

Section type include trapezoid and rectangle.

Section size 1 is the channel height, unit is mm.

Section size 2 is the slope width, unit is mm.

Section size 3 is the bottom width of trapezoidal canal, unit is mm.

For a trapezoidal section, all these three section sizes need to be set, for a rectangular section, only section size 3 needs to be set.

4.5.2 Threshold parameter

The threshold parameters include water level jump threshold, flow velocity jump threshold, etc.

The water level jump threshold and flow velocity jump threshold are mainly used to shield the data jump caused by environmental interference (such as floating objects). When the value exceeds the threshold, corresponding processing is carried out.

Water level jump threshold unit is mm, default value is 50, generally no setup required.

Flow velocity jump threshold unit is cm/s, default value is 30, generally no setup required.

5. Common problems and solutions

5.1 The device cannot be connected to a PLC or computer

Possible reasons:

- 1) The computer has multiple COM ports, and the selected port is incorrect.
- 2) The device address is incorrect, or there are devices with duplicate addresses (factory default is all 1).
- 3) Baud rate, check mode, data bit, stop bit error.
- 4) The host polling interval and waiting time for response are too short and need to be set above 200ms.
- 5) The 485 bus is disconnected, or the A and B wires are connected in reverse.
- 6) If there are too many devices or the wiring is too long, power should be supplied to the nearby area, with a 485 booster and a 120 Ω terminal resistor added.
- 7) USB to 485 driver not installed or damaged.
- 8) Equipment damage.

5.2 Working principle

Radar flow velocity meter adopts the principle of Doppler radar velocity measurement. When measuring the speed of the water surface, the radar velocity sensor emits microwave to the water



surface, and the microwave will be absorbed and reflected when it meets the water surface. Part of the reflected wave is received by the probe of the flow velocity meter, and then converted into an electrical signal, processed by the measurement circuit and measured the Doppler frequency shift, and then according to the above principle to calculate the water flow rate through the signal processing algorithm.

The radar water level meter adopts FMCW modulation mode and uses triangular wave as modulation signal.

The average velocity can be obtained through the radar flow velocity meter, and the water depth can be obtained through the radar water level meter. The sectional flow rate is equal to the average velocity $\times$ cross-sectional area $\times$ correction factor.

After the flow data is obtained, the water volume data within a period of time is equal to the flow $\times$ this period of time.

6. Precautions

1)Warning: Risk of Personal Injury. This device must not be used as a safety device, emergency stop device, or for any other purpose where equipment failure could result in personal injury. Usage Restrictions: For authorized intended use only. The technical manual must be consulted before installation, operation, or maintenance. Failure to comply with the above guidelines may result in death or serious injury.

7. Warranty Statement

The warranty period is 12 months from the date of purchase (subject to valid proof of purchase). During the warranty period, for equipment under normal use and maintenance, if the equipment itself has defects in mechanical parts, materials or craftsmanship that cause malfunctions, and the defect is verified to be true, our company will provide free repair and replacement of parts.

Beyond the warranty period, lifelong maintenance services will be provided.

The following situations are not covered by the warranty:

- (1) Damage to the product caused by incorrect installation, use or operation.
- (2) Disassembly, repair, alteration, modification by non-company technicians, or the user's own replacement of any internal components of the equipment.
- (3) Damage caused by negligent use or the ingress of water or other substances into the equipment.
- (4) Malfunctions or damage resulting from accidents or natural disasters.
- (5) Malfunctions or damage caused by exceeding the operating parameters listed in the product specifications.



8. Contact information

Shandong Renke Control Technology Co., Ltd.

Marketing Center: 10th floor, East Block, No.8 buildings, Shuntai Plaza, High-tech Zone, Jinan City, Shandong Province, China

Post code: 250101

Phone: 400-085-5807

Fax: (86)0531-67805165

Website: www.renkeer.com

Cloud platform address: en.0531yun.com



Shandong Renke Control Technology Co., Ltd

Official website



Welcome to pay attention to the WeChat public platform, enjoy convenient service

9. Document history

- V1.0 Document establishment
- V1.1 Add parameters
- V1.2 Modify operating temperature
- V1.3 Revise register instruction
- V1.4 Add register
- V1.5 Modify parameters
- V1.6 Update appearance
- V1.7 Update register instruction
- V1.8 Add new function
- V1.9 Update installation instruction
- V2.0 Update parameters
- V2.1 Update schematic diagram